

Amir Jamali

LinkedIn | mohammadamirjamali2020@gmail.com | GitHub

Education

Master of Information Technology Engineering (GPA: 4 out of 4)

2017— 2019

Islamic Azad University, Birjand branch

Supervisor: Dr. Sadegh Zeynali

Thesis: Bitcoin price prediction using Bayesian neural networks

Abstract: This study uses Bayesian neural networks and COT data (2018–2019) alongside Bitcoin's weekly prices to predict a cryptocurrency's price with 71% accuracy. Temporal gaps between datasets limit predictions to five days ahead. The method is effective short-term but less so long-term.

Bachelor of Industrial Engineering

2009— 2014

Payam Noor University, Birjand branch

Research Interests

- Software Engineering
- Machine Learning
- Data Science
- Artificial intelligence

Teaching Experiences

Online Instructor I Private Educational Institute

Designed tailored lesson plans and coding exercises for classes of 15–20 students to meet diverse skill levels and enhance practical abilities.

PHP Programming

Jan 2015—Sep 2015

- Taught the fundamentals of PHP, including syntax, functions, and database integration.
- Guided students in developing dynamic web applications using PHP and MySQL.
- Provided hands-on exercises to enhance practical coding skills.

JavaScript Programming

Sep 2015—Feb 2016

- Covered core JavaScript concepts such as DOM manipulation, event handling, and ES6 features.
- Helped students build interactive front-end components for web applications.
- Focused on real-world problem-solving techniques and project-based learning.

C++ Programming

Feb 2016—Nov 2016

- Delivered lessons on C++ basics, object-oriented programming, and data structures.
- Assisted students in implementing algorithms and optimizing code efficiency.
- Conducted coding challenges to strengthen analytical and logical thinking.

Research and Publications Experiences

- An introduction to Quantum Computing; Advantages and challenges – 2021

6th National Conference on Distributed Computing and Big Data Processing - [English version](#)

- Comparison of the local generalization error model (LG Trader) with the exponential moving average (EMA) in selecting stocks with the highest return rate – 2019

Fourth National Conference on Applied Research in Electrical Engineering, Mechanics, Computer and Information Technology

- Predicting Bitcoin price with Bayesian Neural Network by using COT weekly report (Master's thesis) – 2019
[English version](#)
- Building a secure and trustworthy data management framework for explainable machine learning pipelines, incorporating advanced techniques – in progress

English Score

TOEFL : Overall 94

Speaking: 23

Listening: 28

Writing: 22

Reading: 21

Work Experiences

Senior Software Engineer at Pangian I San Francisco - USA

Aug 2023—Feb 2026

Remote community I worldwide Silicon Valley

- **System Architecture:** Designing scalable and efficient system architectures for high-demand platforms.
- **AI Solutions:** designing and enhancing architecture and APIs, integrating advanced AI technologies such as Natural Language Processing (NLP), and Large Language Models (LLMs).
- **DevOps:** Streamlining development with CI/CD pipelines, Docker, and Kubernetes for automated and efficient deployments.

Software Engineer / CTO at Enviroble I London - UK

Mar 2021—Nov 2022

Environmental Social Platform, supported by University College London, BaseKX

- **Data Science:** Utilizing statistical analysis and machine learning techniques to extract insights from complex datasets.
- **Web Development:** Creating responsive and scalable web applications with a focus on usability and performance.
- **System Architecture:** Designing efficient, scalable, and robust system architectures tailored to project needs.

Software Engineer /Founder at Tradeoption I Mashhad - Iran

Jan 2020—Feb 2022

Algorithmic trading platform to contribute to Stock Options Contracts Trading.

- **Algorithm Design:** Developing optimized and innovative algorithms for solving real-world business challenges.
- **Web Development:** Building efficient and user-focused web solutions using modern frameworks.
- **Data Science:** Applying predictive models and analytical techniques to drive data-driven decisions.

Software Developer/Co-Founder at Class724 I Birjand - Iran

May 2015—Nov 2019

Learning Management System (LMS) to facilitate the teaching of programming skills.

Science and Technology Park of South Khorasan

- **Web Development:** Designing and maintaining interactive and functional websites with scalable architectures.
- **System Design:** Structuring and implementing reliable systems with a focus on operational efficiency.
- **DevOps:** Automating and enhancing development workflows using tools like Docker, Jenkins, and Kubernetes.

Honors

- Involved in finalists of UCL Hatchery Social Enterprise Award with Enviroble.Ltd (Santander Awards 2022)

- Ranked as the second-highest GPA among MSc students in the Computer Science faculty, with a GPA of 18.04/20

Courses

- | | |
|--|--------------------------------|
| • Network ++ | Tehran Institute of Technology |
| • Active Directory | Tehran Institute of Technology |
| • Network Infrastructure | Tehran Institute of Technology |
| • Windows server Enterprise Administration | Tehran Institute of Technology |

Skills

Technical Skills

- | | | |
|------------------------|-----------------|---------------------------------------|
| • Machine Learning | • DevOps | • Microsoft Certified IT Professional |
| • Software Engineering | • System Design | |

Programming Languages & Frameworks

- | | | |
|-----------------------------------|--------------|---|
| • Python, Django, Flask, Fast API | • Node.js | • Next.js |
| • PHP (full stack WordPress) | • React | • Flask |
| • Java | • Django | • TensorFlow, Pytorch, ScikitLearn, PySpark, Qiskit |
| • C++, MQL | • Angular | • E2E Testing (Playwright, Automated QA in CI/CD) |
| • C | • TypeScript | |
| • JavaScript | • MQL | |

Databases & Tools

- | | | |
|--------------|-----------------|--------------|
| • MySQL | • Elasticsearch | • Docker |
| • Redis | • Apache | • Databricks |
| • MongoDB | • Kafka | |
| • PostgreSQL | • Linux | |
| • SQLite | • Nginx | |

Embedded Systems & Microcontroller

Microprocessor & Assembly Programming (EMU8086, AVR)

- Microprocessor structure and components
- 8086 & Pentium architecture

ARM Microcontroller Training with Keil (LPC1768)

- ARM microcontroller families
- LPC1768 initialization

- Assembly programming with EMU8086
- 8086 CPU design
- USART registers in AVR microcontrollers

- GPIO, NVIC, ADC, Timer, PWM, DAC, USART, RTC modules